

# What Survives Transport

## Second-Order and Global Invariants of Resolution Geometry: the Physical Sector, the Geometric Phase, and Independence Results

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### Abstract

We probe resolution geometry by transport—pushing quantities around loops and through changes of representation—and ask which structures survive. The organizing result is a clean split of the framework into a transport-invariant *physical sector* and a transport-carried *geometric-phase sector*. The physical sector—entropy production, the resolved-null coupling, and the reactive Lorentzian signature—is holonomy-invariant, each quantity protected by a distinct established theorem (scalar invariance; the structure group is the coupling’s stabilizer; Sylvester’s law of inertia). The global data that these invariants cannot see is instead carried by the reactive time-axis, whose rotation around a loop equals the connection’s holonomy: a geometric phase of Berry–Hannay type, which supplies an observable handle on the otherwise abstract global classification. Around this centerpiece we establish several sharp independence and structure results, each numerically verified: the genuinely geometric second-order object is the observation-space reactive frame, not the parameter-space Hessian or Fisher metric (which are coordinate representations); Fisher information (local) and holonomy (global) are independent invariants, neither determining the other; the similarity type, automorphism group, and holonomy are conformally invariant while Fisher carries units, so the geometry is naturally conformal; identical first-order dynamics with different hidden null-sector memory produce different observable couplings, so observable geometry encodes hidden memory; and discrete plaquette transport converges at second order to the continuous path-ordered holonomy, bridging the atlas to differential geometry. We report the results with an explicit gradation of strength and mark the boundaries: the constructions are observation geometry, and their kinship to gauge theory is formal.

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## 1 Method: probe by transport

A structure is geometric to the extent that it survives transport—around loops (holonomy), under change of parametrization (gauge), and under refinement (discretization). This paper applies that probe systematically. The payoff is a split of resolution geometry into what transport leaves invariant (the physical sector) and what it carries (the global, geometric-phase sector), together with a set of independence results that sharpen “the framework sees more than Fisher” into precise statements. Every claim is numerically verified; we grade each by strength and state the boundaries plainly.

## 2 The physical sector is holonomy-invariant

Let the structure group be  $\text{Aut}(\mathcal{G})$  (the compact coupling-stabilizer group), and let a connection have loop holonomy  $H \in \text{Aut}(\mathcal{G})$ .

**Theorem 2.1** (Three protection theorems). *The following are invariant under holonomy transport by any  $H \in \text{Aut}(\mathcal{G})$ :*

1. *the entropy production  $\sigma_{\text{EP}} = \text{tr}(\nu^\top D^{-1} \nu \Sigma)$ , because it is a scalar and  $H$  acts by orthogonal conjugation;*
2. *the resolved-null coupling  $C$ , because  $\text{Aut}(\mathcal{G})$  is by definition its stabilizer,  $ACB^\top = C$ ;*
3. *the reactive signature  $\text{sig}(K - \omega^2 M)$ , because  $H$  acts by congruence and congruence preserves inertia (Sylvester’s law).*

*Proof.* (1) Under  $y \mapsto Hy$  the drift, diffusion, and covariance conjugate by  $H$ , so  $\nu \mapsto H\nu H^\top$  and  $\sigma_{\text{EP}} \mapsto \text{tr}(H\nu^\top D^{-1} \nu \Sigma H^\top) = \sigma_{\text{EP}}$  by cyclicity and  $H^\top H = I$ . (2) Immediate from the defining relation of  $\text{Aut}(\mathcal{G})$ . (3) Congruence  $X \mapsto HXH^\top$  preserves the numbers of positive, negative, and zero eigenvalues (Sylvester). Verified: loop-commutators  $\sim 10^{-16}$  for all three.  $\square$

*Remark 2.2* (Why this is the right kind of robustness). These are exactly the invariants of the compact structure group, and invariants of a compact group are holonomy-blind. The physical content of the framework thus descends to the base as genuine global invariants, not gauge artifacts. This is the honest answer to “do the physical values survive higher-order transport?": yes, by three separate theorems.

## 3 The geometric phase of the reactive time-axis

Precisely because they are invariant, the quantities of Theorem 2.1 cannot detect the holonomy. The global datum is carried elsewhere.

**Theorem 3.1** (Reactive time-axis as geometric phase). *The distinguished eigendirection of the reactive response form (the timelike axis, the negative-eigenvalue direction of  $K - \omega^2 M$  on the Lorentzian window) is not holonomy-invariant: transported around a loop with holonomy  $H \in \text{Aut}(\mathcal{G})$  it returns as  $Hv$ . Its loop rotation equals the holonomy and is a geometric phase (Berry–Hannay type).*

*Proof.* The transported form is  $H(K - \omega^2 M)H^\top$ , whose timelike eigenvector is  $Hv$ ; the angle between  $v$  and  $Hv$  is the holonomy angle. Verified for  $r = 2$ : holonomy angle 0.700 rad reproduced as the time-axis rotation to  $10^{-3}$ .  $\square$

**Corollary 3.2** (Observable handle on the global classification). *The global datum (the monodromy  $\rho : \pi_1(B) \rightarrow \text{Aut}(\mathcal{G})$ ) is invisible to the physical invariants but visible as the geometric phase of the reactive time-axis. Measuring these phases around a generating set of loops determines  $\rho$  up to conjugacy—an operational route to the system’s point in the character variety.*

*Non-claim 3.3.* This localizes and operationalizes the global classification; it does not compute the character variety as a space, and the phase is that of the resolution-geometry connection (formal Berry–Hannay), not a physical quantum phase.

## 4 The geometric second-order object

**Proposition 4.1** (Observation-space frame vs. parameter-space Hessian). *Under a reparametrization  $\theta \mapsto \phi$  (which leaves the first-order response subspace and the observation metric fixed): the resolved subspace  $R$ , null sector  $N$ , covariance  $\Sigma$ , canonical correlations, and the reactive frame are invariant; the parameter-space Fisher metric and raw Hessian transform (they are coordinate representations, the Hessian carrying an inhomogeneous connection term). Hence the genuinely geometric second-order object lives on observation space, not on parameter space.*

*Proof.*  $\text{Im}(JM^{-1}) = \text{Im}(J)$  for invertible  $M$ , so  $R$  (and  $N, \Sigma$ ) are fixed; the reactive frame lives on observation space, untouched by parameter change. The Fisher metric transforms as  $M^{-\top}(\cdot)M^{-1}$  and the raw Hessian acquires Christoffel terms. Verified:  $|P_R - P'_R| \sim 10^{-16}$  while  $|F - F'| = 1.18$ ,  $|H - H'| = 6.44$ .  $\square$

*Remark 4.2* (Strength). This is sharp conceptually but follows from where each object lives; we present it as a clarification (the second-order geometric content is the observation-space frame) rather than a deep theorem.

## 5 Independence of Fisher and holonomy

**Proposition 5.1** (Fisher  $\perp$  holonomy). *Fisher information (a local invariant) and holonomy (a global invariant) are independent: there exist geometries with equal Fisher and different holonomy, and with equal holonomy and different Fisher. Neither determines the other.*

*Proof.* Fisher  $J^\top \Sigma^{-1} J$  is computed at a chart; holonomy is a product of transitions around a loop. Fixing the chart and varying the transition cocycle gives equal Fisher, different holonomy; fixing the cocycle and varying the chart Jacobian gives equal holonomy, different Fisher. Verified: equal Fisher with  $|H_1 - H_2| = 0.84$ ; equal holonomy with  $|F_c - F_d| = 13.3$ .  $\square$

*Remark 5.2* (Strength). The independence is that of a local versus a global invariant, hence expected; its value is the precise statement, which upgrades “sees more than Fisher” into “Fisher and holonomy classify logically independent data.”

## 6 The geometry is conformal

**Proposition 6.1** (Conformal invariance). *Under a global rescaling  $\Sigma \mapsto \lambda \Sigma$  ( $\lambda > 0$ ): the canonical correlations, the automorphism group  $\text{Aut}(\mathcal{G})$ , and the holonomy are invariant, while the Fisher information scales as  $\lambda^{-1}$ . Hence the similarity type is a conformal invariant and the natural protocol for resolution geometry is conformal, not strict.*

*Proof.* The whitened coupling  $\Sigma_R^{-1/2} C \Sigma_N^{-1/2}$  is unchanged under  $\Sigma \mapsto \lambda \Sigma$ , so canonical correlations are invariant;  $O(\lambda \Sigma_\bullet) = O(\Sigma_\bullet)$  and the coupling constraint is scale-free, so  $\text{Aut}(\mathcal{G})$  and holonomy are unchanged;  $J^\top (\lambda \Sigma)^{-1} J = \lambda^{-1} J^\top \Sigma^{-1} J$ . Verified over 100 values of  $\lambda \in [10^{-2}, 10^2]$ : canonical-correlation deviation  $\leq 6 \times 10^{-17}$ ; correlation of Fisher trace with  $\lambda^{-1}$  equal to 1.000000.  $\square$

## 7 Hidden memory is observable

**Proposition 7.1** (Observable geometry encodes null-sector memory). *Two systems with identical first-order resolved dynamics and identical instantaneous cross-drift, but different null-sector dynamics (hence different Mori–Zwanzig memory kernels  $K(\tau) = A_{RN}e^{A_{NN}\tau}A_{NR}$ ), have different stationary resolved–null couplings. The resolution geometry therefore distinguishes systems that agree at first order but differ in hidden memory.*

*Proof.* The stationary covariance solves the Lyapunov equation, whose off-diagonal block depends on  $A_{NN}$  through the kernel; changing  $A_{NN}$  changes  $C$ . Verified: identical  $A_{RR}$  and cross-drift, null relaxation changed, memory-kernel weight  $0.449 \rightarrow 1.797$  and  $|C_1 - C_2| = 28.7$ .  $\square$

*Remark 7.2* (Strength). This is one of the sharper results: it is a genuine, physics-flavored statement (observable geometry sees hidden dynamical memory), and it rests on the proved Mori–Zwanzig structure rather than on representation.

## 8 Discrete transport converges to a differential-geometric connection

**Proposition 8.1** (Convergence of plaquette holonomy). *For a smooth non-abelian connection, the discrete holonomy computed by an  $n$ -segment product converges to the continuous path-ordered holonomy as  $n \rightarrow \infty$ , at second order in the mesh ( $O(n^{-2})$  for the midpoint rule). The atlas connection is thus the discretization of a genuine differential-geometric (Ehresmann) connection.*

*Numerical confirmation.* For a direction-varying  $\mathfrak{so}(3)$  connection the error  $\|H_n - H_\infty\|$  falls by a factor  $\approx 4$  per doubling of  $n$  ( $6.7 \times 10^{-1}, 1.5 \times 10^{-1}, 3.5 \times 10^{-2}, \dots, 3.4 \times 10^{-5}$  for  $n = 2, \dots, 256$ ), the signature of  $O(n^{-2})$  convergence to the product integral.  $\square$

## 9 Supporting results

**Proposition 9.1** (Curvature versus nonlinearity). *Holonomy is a property of the connection (transport), not of dynamical nonlinearity. A highly nonlinear system can have zero holonomy and a linear one large holonomy: the two are independently assignable. Verified: system A (excess kurtosis  $-0.34$ , holonomy  $0$ ), system B (kurtosis  $0$ , holonomy  $1.23$ ), system C (both).*

**Proposition 9.2** (Defect localization). *A single inadmissible edge renders non-admissible only the plaquettes that contain it. Verified: corrupting one horizontal edge on a lattice produces exactly two non-admissible plaquettes (its two neighbors); the defect is local.*

**Proposition 9.3** (Constructive repair). *The nearest admissible transition to an inadmissible  $L$  is the polar projection of  $L$  onto the covariance-isometry group (in the whitened frame). Verified: a transition of residual  $4.40$  is repaired to residual  $1.4 \times 10^{-15}$  at minimal Frobenius change  $2.17$ . The framework is thereby constructive: it returns the nearest admissible atlas rather than only a verdict. Global coherence repair is an optimization problem and remains open.*

## 10 Scope and boundaries

**Strength gradation.** The sharpest results are the geometric-phase identification (Thm 3.1, Cor 3.2), the memory observability (Prop 7.1), the conformal invariance (Prop 6.1), and the convergence to a continuous connection (Prop 8.1). The independence, geometric-object, curvature-versus-nonlinearity, and localization results (Props 4.1, 5.1, 9.1, 9.2) are true and precise but partly structural (local versus global, or “different objects”); they are presented as clarifications.

**Not run: universality.** A universality claim—that unrelated systems fall into few transport classes—cannot be established synthetically, since the framework is domain-agnostic by construction; it requires real datasets and is deferred to empirical work.

**No physics beyond formal kinship.** The connection, curvature, holonomy, and geometric phase are those of an observation geometry. Their formal identity with gauge-theoretic objects is resemblance, not a claim of a physical gauge field, spacetime, or field dynamics.

## 11 Conclusion

Probing resolution geometry by transport splits it cleanly. The physical sector—entropy production, coupling, and reactive signature—is holonomy-invariant, protected by three separate theorems, and so is genuinely global. The global data these invariants cannot see is carried by the reactive time-axis as a geometric phase, which turns the abstract global classification into an observable and supplies a reconstruction route. Around this, transport probes yield sharp statements: the geometric second-order object is the observation-space frame; Fisher and holonomy are independent; the geometry is conformal; hidden memory is observable; and discrete transport converges to a genuine differential-geometric connection. Reported with an honest gradation and clear boundaries, these are the second-order and global invariants of the framework—what survives transport, and what transport carries.

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